Long run problems

1) Find the unique fixed probability vector t of

$$P = \begin{bmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Soln:- Let $t = (x, y, z)$ be the fixed probability vector. Then $x+y+z=1$

$\Rightarrow z = 1-x-y$

Since t is prob vector,

$tP = t$

$$(x, y, 1-x-y) \begin{bmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{bmatrix} = (x, y, 1-x-y)$$

$\Rightarrow \frac{1}{2}y = x \Rightarrow y = 2x \Rightarrow 2x - y = 0$ —— (1)

$\frac{3}{4}x + \frac{1}{2}y + 1 - x - y = y \Rightarrow -\frac{1}{4}x - \frac{3}{2}y = -1$

$\Rightarrow x + 6y = 4$ —— (2)

$\frac{1}{4}x = 1 - x - y \Rightarrow \frac{5}{4}x + y = 1$

$\Rightarrow 5x + 4y = 4$ —— (3)

On solving $x = \frac{4}{13}$, $y = \frac{8}{13}$, $z = \frac{1}{13}$

$\therefore t = \left(\frac{4}{13}, \frac{8}{13}, \frac{1}{13}\right)$

2) Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for a Santro. However, if he has a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2000, he bought his first car which was a Santro.

(i) Find probability that he has

a) 2002 Santro

b) 2002 Maruti

c) 2003 Ambassador

d) 2003 Santro

(ii) In the long run, how often will he have a Santro?

Solⁿ:- Let a_1 be the state of having Maruti
Let a_2 be the state of having Ambassador
Let a_3 be the state of having Santro.

In 2000 he has Santro. Therefore, for 2001,

$$P = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \end{array}$$

For 2002,

$$P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

$$= \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ \boxed{1/9} & 4/9 & \boxed{4/9} \end{bmatrix} \end{array}$$

Santro 2000 & 2002

In 2000 Santro & 2002 Maruti

∴ a) Probability that he has Santro in 2002 is 4/9 & b) Maruti is 1/9.

For 2003,

$$P^3 = \begin{bmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

$$P^3 = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \\ \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \\ 4/27 & \boxed{7/27} & \boxed{16/27} \end{bmatrix} \end{array}$$

2000 Santro & 2003 Santro

2000 Santro & 2003 Ambassador

$\therefore$ c) Probability that he has Ambassador in 2003 is $7/27$ & d) Santro is $16/27$.

(ii) For long run,

$$(x \quad y \quad 1-x-y) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} = (x \quad y \quad 1-x-y)$$

$$1-x-y = 3x$$
$$3x + (1-x-y) = 3y$$
$$3y + (1-x-y) = 3(1-x-y)$$

On solving, we get $y = 1/3$, $x = 1/6$, $z = 1/2$

$$t = \left(1/6,\ 1/3,\ 1/2\right)$$

$\therefore$ In the long run, he has Santro 50% of the time.

3) For a Markov chain, the transition matrix is

$$P = \begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix}$$

with initial distribution $p^{(0)} = \left(1/4 \quad 3/4\right)$.

Find

(i) $P_{21}^{(2)}$ (ii) $P_{12}^{(2)}$ (iii) $p^{(2)}$

(iv) the vector $p^{(0)} P^n$ approaches

(v) P^n approaches.

$$P^2 = \begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix} = \begin{bmatrix} 5/8 & 3/8 \\ 9/16 & 7/16 \end{bmatrix}$$

(i) $P_{21}^{(2)} = 9/16$

(ii) $P_{12}^{(2)} = 3/8$

(iii) $p^{(2)} = p^{(0)} P^2 = \left(1/4 \quad 3/4\right) \begin{bmatrix} 5/8 & 3/8 \\ 9/16 & 7/16 \end{bmatrix}$

$$= \left(\frac{37}{64}, \frac{27}{64}\right)$$

(iv) $p^{(0)} P^n$ approaches the fixed unique probability vector $t = (x, y) = (x, 1-x)$

$$(x \quad 1-x)\begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix} = (x \quad 1-x)$$

$$\Rightarrow \frac{1}{2}x + \frac{3}{4} - \frac{3}{4}x = x$$

$$\Rightarrow x = \frac{3}{5} \quad \& \quad y = 1-x = 1-\frac{3}{5} = \frac{2}{5}$$

$$\therefore t = \left(\frac{3}{5}, \frac{2}{5}\right)$$

(v) $$P^n = \begin{pmatrix} t \\ t \end{pmatrix} = \begin{pmatrix} 3/5 & 2/5 \\ 3/5 & 2/5 \end{pmatrix}$$

4) A salesman's territory consists of 3 cities A, B and C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. However, if he sells in either B or C, then the next day he is twice as likely to sell in city A as in other city. In the long run, how often does he sell in each of the cities?

Solⁿ:- A : he sells in city A

B : he sells in city B

C : he sells in city C

$$P = \begin{array}{c|ccc} & A & B & C \\ \hline A & 0 & 1 & 0 \\ B & 2/3 & 0 & 1/3 \\ C & 2/3 & 1/3 & 0 \end{array}$$

For long run,

$$(x \quad y \quad 1-x-y) \begin{bmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix} = (x \quad y \quad 1-x-y)$$

On solving,

$x = 2/5 \qquad y = 9/20 \qquad z = 3/20$

$t = \left(2/5, \ 9/20, \ 3/20 \right)$

$\therefore$ In the long run, he sells 40% of the times in city A, 45% in city B & 15% in city C.

5) In a certain market, only 2 brands of cold drinks A & B are sold. Given that a man's last purchased brand A, there is 80% chance that he would buy same brand in the next purchase. While if the man purchased brand B, there is 90% chance that his next purchase would be brand B. Using this transformation,

(i) develop TPM

(ii) Determine the long - run loyalty for brand A and brand B.

(i)
$$\begin{array}{c|cc} & A & B \\ \hline A & 0.8 & 0.2 \\ B & 0.1 & 0.9 \end{array}$$

(ii)
$$(x \quad 1-x)\begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix} = (x \quad 1-x)$$

$$0.8x + 0.1(1-x) = x$$
$$\Rightarrow 0.8x + 0.1 - 0.1x = x$$
$$\Rightarrow x + 0.1x - 0.8x = 0.1$$
$$\Rightarrow 0.3x = 0.1$$
$$\Rightarrow x = 1/3 \quad \Rightarrow y = 1-x = 1-1/3 = 2/3$$

$$\therefore t = \left(1/3, 2/3\right)$$

$\therefore$ In the long run he purchases 33.33% of the time brand A and 66.67 of the time brand B.